2043 Spring 生化学 2 English, Home work#4(250407) dead line: 23:59, April 14, 2025

学号:

姓名:

Q1. 判断题（请在括号里画√或 X, 1 分/题, 共 30 分）

(1) Lavoisier said that animals somehow transform chemical fuels (foods) into heat. ( )
(2) First law of thermodynamics is the principle of the conservation of energyis. ( )
(3) Second law of thermodynamics states that the universe always tends toward deceasing disorder. ( )
(4) If ΔG is positive, it is exergonic. ( )
(5) Enthalpy (H) is the heat content of the reacting system. ( )
(6) Entropy (S) is the quantitative expression for the randomness or disorder in a system. ( )
(7) 1 cal = 4.184J. ( )
(8) In standard transformed constants, the standard state is defined at pH 0. ( )
(9) In standard transformed constants, the Mg2+ concentration is set to 10 mM. ( )
(10) If K'_{eq} is 10^2, the $\Delta G'^{O}$ is -11.4kcal/mol. ( )
(11) Given that, starting with 10 mM glucose 1-phosphate and no glucose 6-phosphate, the final equilibrium mixture at 25 ^{o}C and pH 7.0 contains 1.0 mM glucose 1-phosphate. In this situation, $\Delta G'$ is roughly -5.4kJ/mol. ( )
(12) If the standard free-energy change $\Delta G'^{O}$ is negative, the actual free-energy change is always negative. ( )
(13) In the homolytic cleavage, electrons are divided to same amount. ( )
(14) Cleavage of a C-H bond to Carbanion + Proton is homolytic cleavage. ( )
(15) In the transfer of phophoryl group from ATP to Glucose, AMP is the leaving group. ( )
(16) Hydrolysis of phosphoenolpyruvate (PEP) makes Pyruvate as the product, where tautomerization converts between enol- and keto-form. ( )
(17) Because of the extra resonance stabilization, DG of thioester hydrolysis is large. ( )
(18) In Yeast Inorganic polyphosphate (polyP) exist in high concentraion like 200 mM. ( )
(19) Oxidized is gain of electrons. ( )

(20) Reducing agent oxidized the target, while itself is reduced. ( )

(21) Oxidation and reduction must occur together. ( )

(22) An electron donor and corresponding acceptor constitute a conjugate redox pair. ( )

(23) In many cases, oxidation is dehydrogenation. ( )

(24) Biological oxidations are regarded as two reducing equivalents passing from substrate to oxygen. ( )

(25) A half-cell that takes electrons from the reference hydrogen cell is assigned a positive value of E0, and one that donates electrons, a negative value. ( )

(26) Standard reduction potentials can be used to calculate free-energy change using Nernst equation. ( )

(27) NAD+ and NADP+ has similar UV absorption spectra. ( )

(28) NAD binds to the Rossman fold by covalent binding.. ( )

(29) Concentration of NAD+ is higher than that of NADH. ( )

(30) Concentration of NADP+ is higher than that of NADPH. ( )

Q2. Please write down the 4 reasons of why ATP hydrolysis is large and negative。(各 2 分、共 8 分)

Q3. Please write down the 4 methods to transfer Electron (各 2 分、共 8 分)

Q4. Calculate the ΔG of the following, when [Oxaloacetate] and [NADH] are 1.0 M and [Malate] and [NAD] are 0.1M (共 4 分)

$$\text{Oxaloacetate} + \text{NADH} + H^+ \rightarrow \text{Malate} + NAD^+$$